How to read graphs?

- Make and interpret a scatter diagram
- Identify linear and nonlinear relationships and relationships that have a maximum and a Minimum
- Define and calculate the slope of a line
- Graph relationships among more than two variables

Construction of a graph

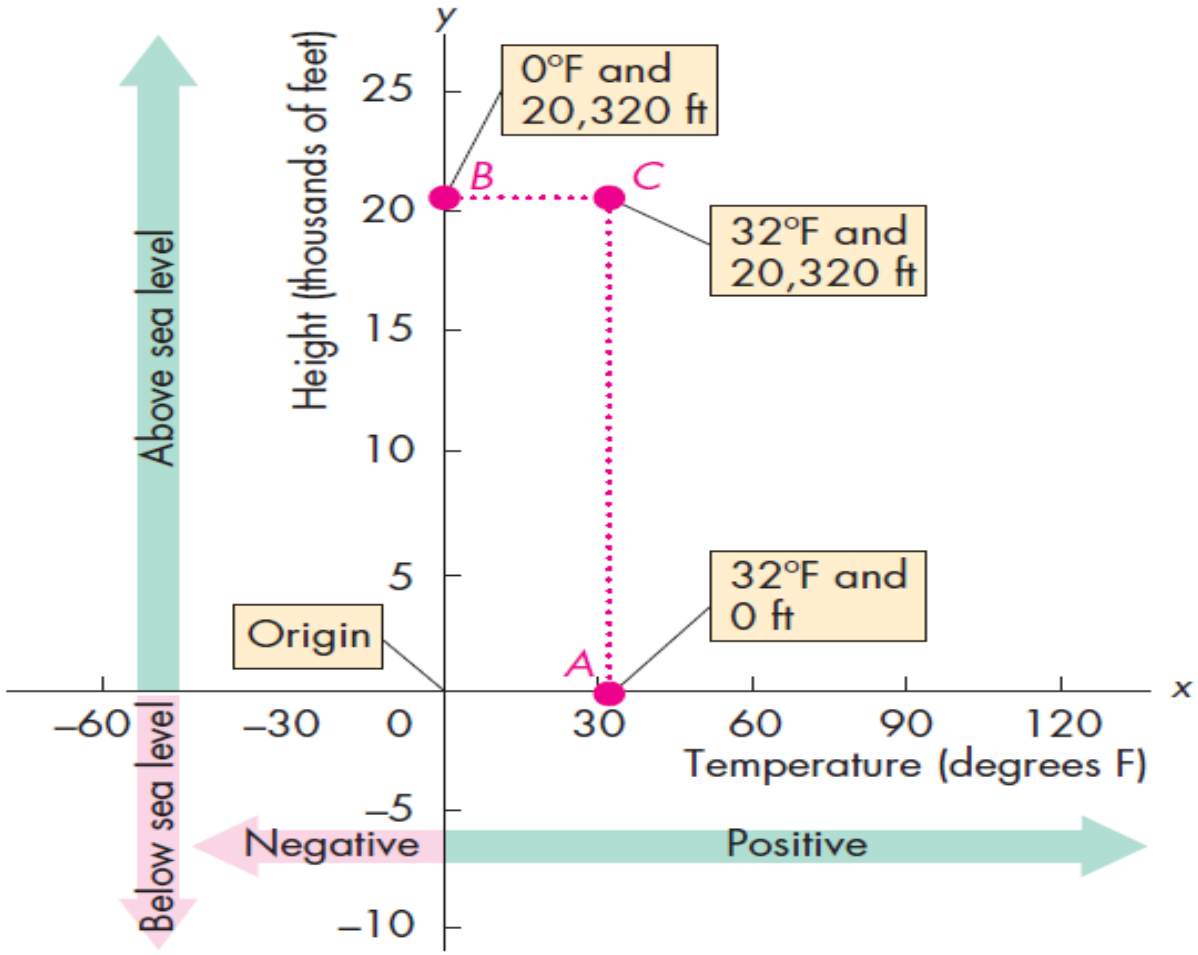
What is a Graph?

A visual representation of the relationship between two variables.

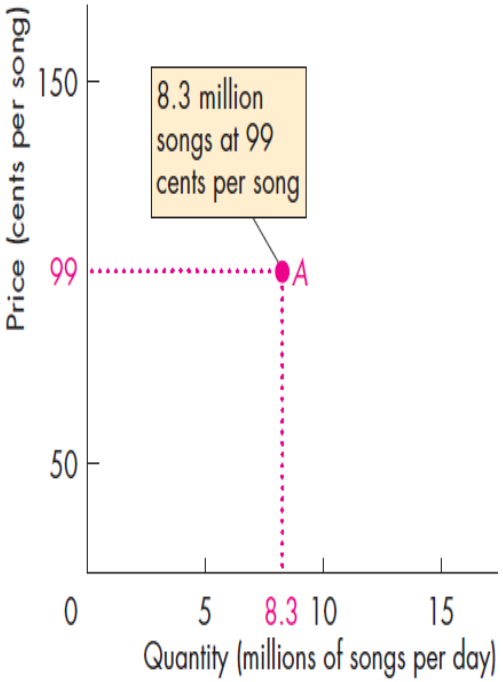
The important points to understand about a graph are:

- What is on each of the two axes (**horizontal and vertical**)?
- What are the units on each axis? What kind of
- The relationship is depicted in the curve or curves shown in the graph?
- The relationship between the two variables in a curve is given by its slope.

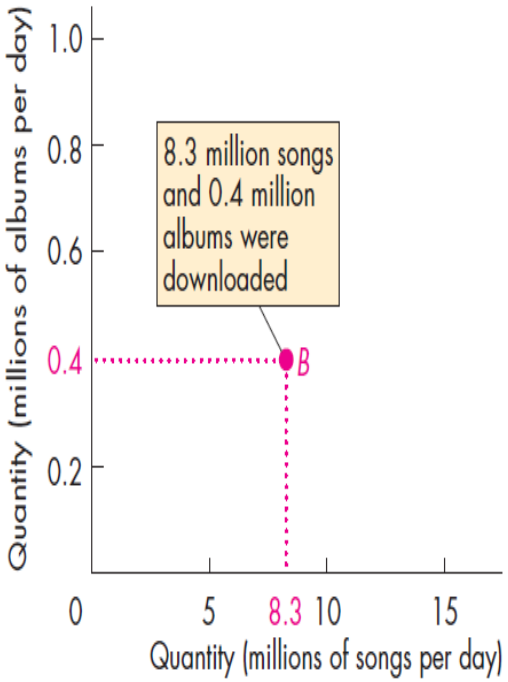
FIGURE A1.1 Making a Graph



Two Graphs of Economic Data



(a) iTunes downloads: quantity and price



(b) iTunes downloads: songs and albums

The graph in part (a) tells us that in January 2010, 8.3 million songs per day were downloaded from the iTunes store at a price of 99 cents a song.

The graph in part (b) tells us that in January 2010, 8.3 million songs per day and 0.4 million albums per day were downloaded from the iTunes store.

Different types of curves

The patterns to look for in graphs are the four cases in which

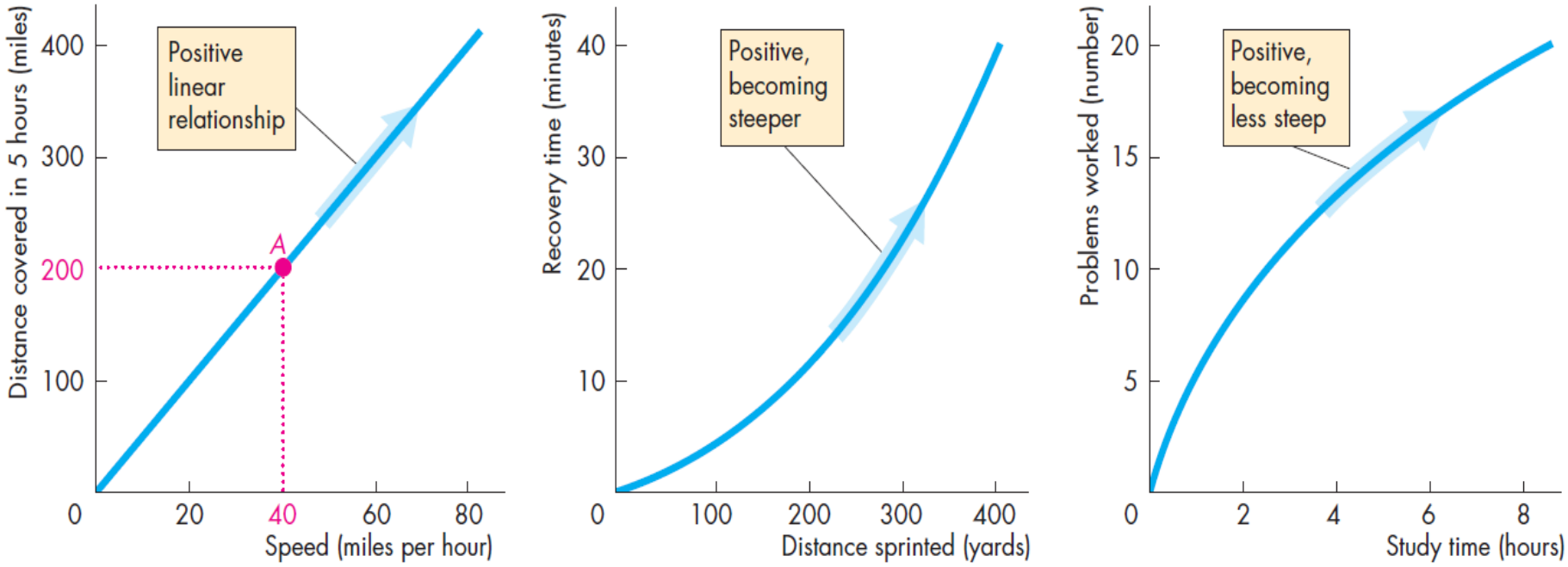
- Variables move in the same direction.
- Variables move in opposite directions.
- Variables have a maximum or a minimum.
- Variables are unrelated.

Let’s look at these four cases.

Variables move in the same direction

- A relationship between two variables moving in the same direction is called a positive or direct relationship.
- A line that slopes upward shows a positive or direct relation.
- A relationship shown by a straight line is called a linear relationship.

Positive or Direct Relationship

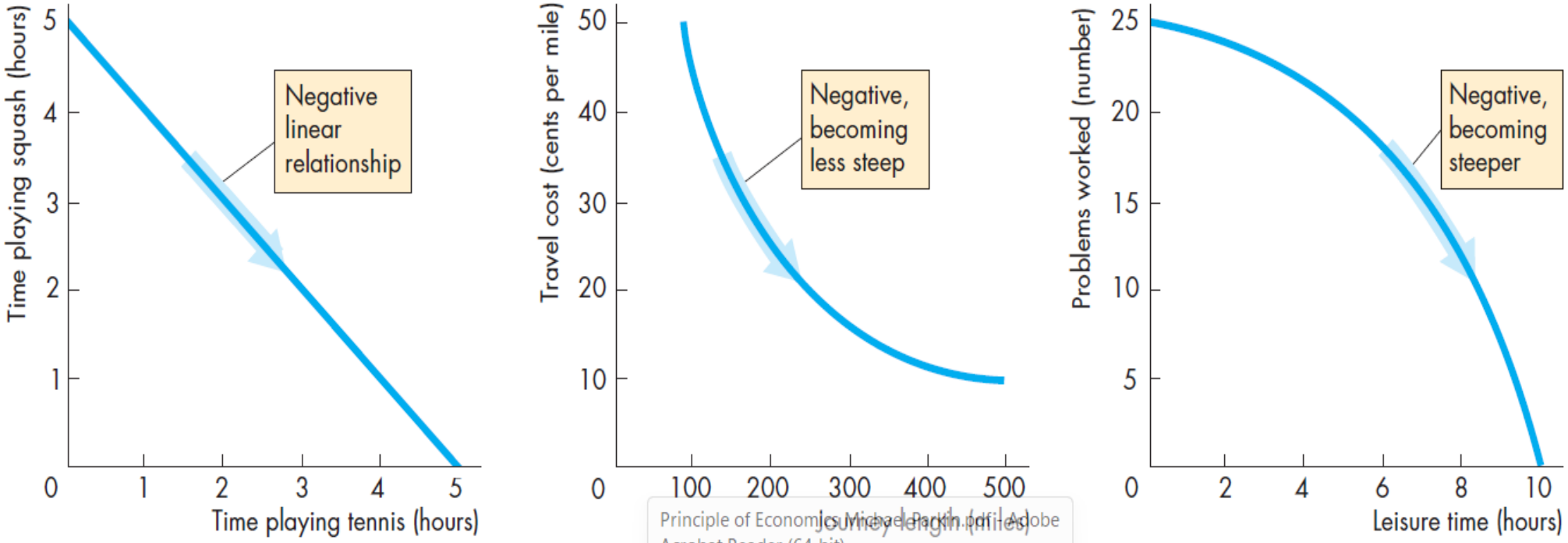


(a) Positive linear relationship **(b) Positive, becoming steeper** **(c) Positive, becoming less steep**

Variables move in opposite directions

- A relationship between variables that move in opposite directions is called a negative relationship or an inverse relationship.
- A line that slopes downward shows a negative or inverse relation.
- A relationship shown by a straight line is called a linear relationship.

Negative or Inverse relationship



(a) Negative linear relationship (b) Negative, becoming less steep (c) Negative, becoming steeper

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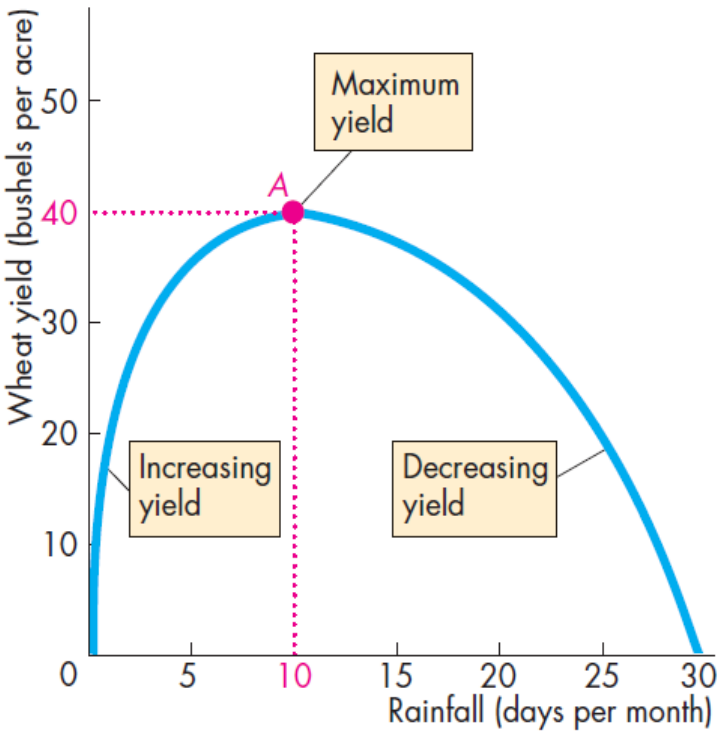
Variables That Have a Maximum or a Minimum

- Many relationships in economic models have a maximum or a minimum. For example, firms try to make the maximum possible profit and to produce at the lowest possible cost.

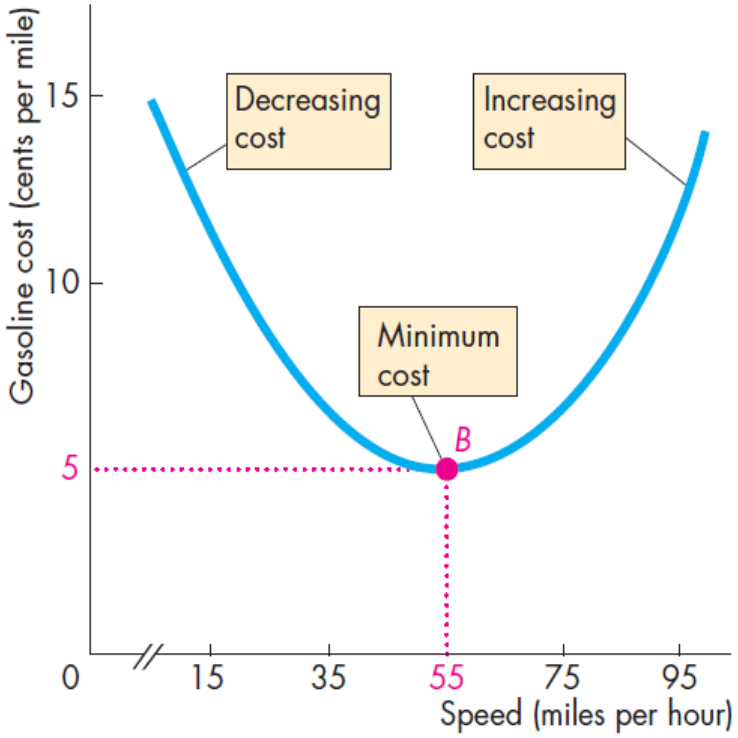
- Either this relationship is one that starts out sloping upward, reaches a maximum, and then slopes downward.

- Or, This relationship is one that starts out sloping downward, reaches a minimum, and then slopes upward.

Maximum and Minimum Points



(a) Relationship with a maximum



(b) Relationship with a minimum

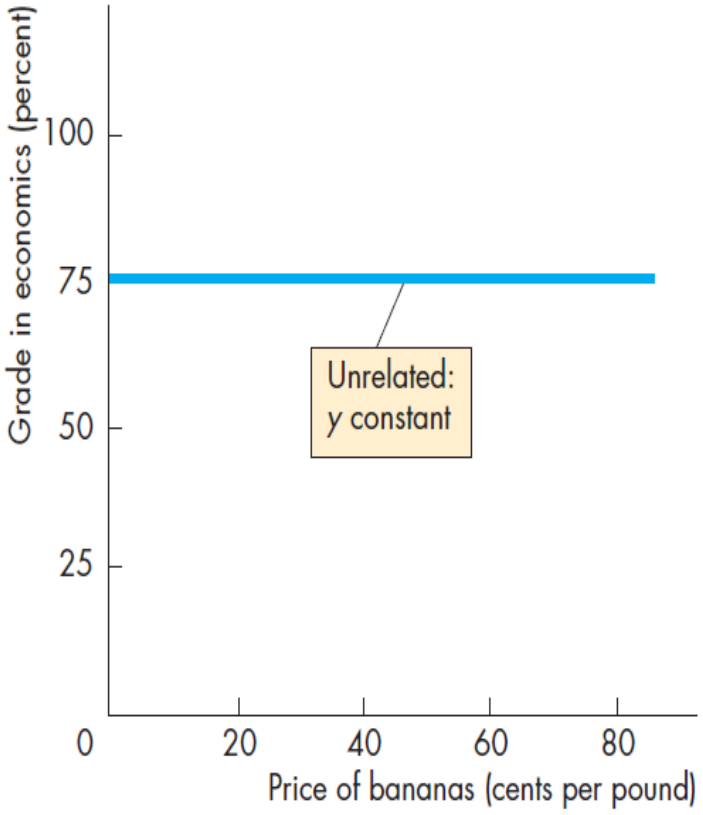
Part (a) shows a relationship that has a maximum point, *A*. The curve slopes upward as it rises to its maximum point, is flat at its maximum, and then slopes downward.

Part (b) shows a relationship with a minimum point, *B*. The curve slopes downward as it falls to its minimum, is flat at its minimum, and then slopes upward.

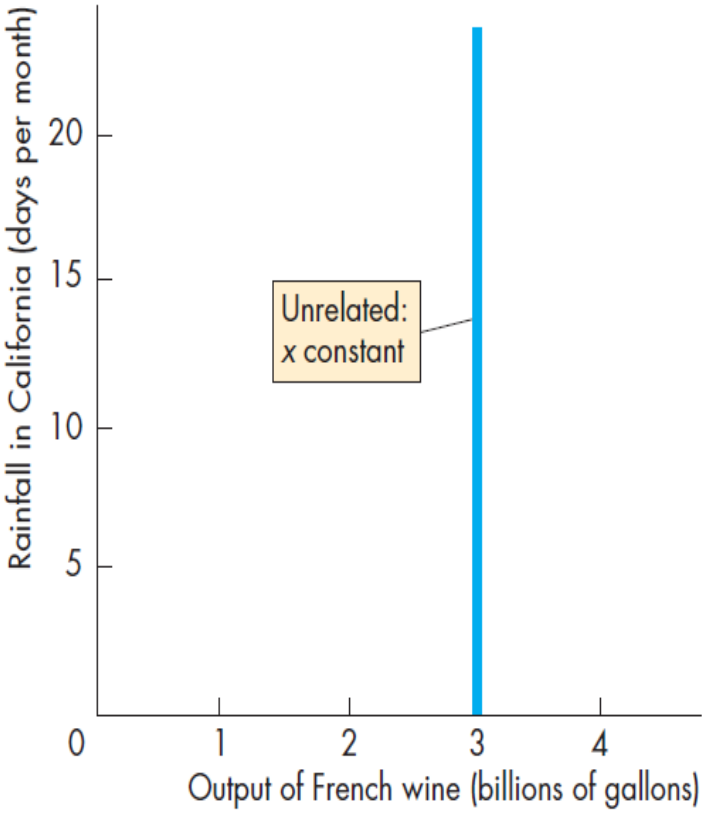
Variables That Are Unrelated

- There are many situations in which no matter what happens to the value of one variable, the other variable remains constant.
- Sometimes we want to show the independence between two variables in a graph, and there are two ways of achieving this.

Variables That Are Unrelated



(a) Unrelated: *y* constant



(b) Unrelated: *x* constant

This figure shows how we can graph two variables that are unrelated. In part (a), a student's grade in economics is plotted at 75 percent on the *y*-axis regardless of the price of bananas on the *x*-axis. The curve is horizontal.

In part (b), the output of the vineyards of France on the *x*-axis does not vary with the rainfall in California on the *y*-axis. The curve is vertical.